

Block**3****ICT: APPROACHES AND APPLICATION**

UNIT 9**ICT for Education**

UNIT 10**ICT for Health**

UNIT 11**ICT for Disability**



UNIT 9 ICT FOR EDUCATION

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9.0 INTRODUCTION

Information and communication technology (ICT) is playing a significant role in every field of life, including education. Modern ICT tools have brought about significant changes in the education sector at all levels, from school education to higher education. You can easily experience the impact of ICT on education. Traditional classrooms are being converted into smart classrooms; time and space-related barriers are being removed, and the concept of learning anytime and anywhere is getting more and more popular.

Now an ocean of knowledge is available for you on the internet. You can get and submit your home assignments online. A number of online courses developed by the institutions and experts of different countries are available for you. You can watch many interesting audiovisual programmes related to your subject topics on the internet. You can do many practical activities through virtual labs. You can watch the lectures of the world's best faculties and interact with them sitting in your room or workplace situated in a small town. You can get access to a massive stock of knowledge completely free of cost available as open educational resources (OER), and a lot more.

Education is an essential indicator of development and 'education for all' is quite crucial for inclusive development of our society. Do you think ICT can help to make the quality education available to all? In this Unit, we will discuss the different aspects of ICT for education.

9.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- describe the scope of ICT in education at different levels;
- explain the primary requirements for successful integration of ICT in our educational system;
- describe the current Indian scenario of ICT integration in education; and
- critically discuss the issues and challenges in the successful integration of ICT in the Indian education system.

9.2 SCOPE OF ICT IN EDUCATION

We can divide the whole educational transactions into two major parts: (i) teaching-learning activities and (ii) administrative activities or management of educational institutions. ICT plays a crucial role on both fronts. We shall discuss the scope of ICT in both types of educational transactions in coming sub-sections.

9.2.1 ICT For Teaching-Learning Process

In this subsection, we shall discuss the scope of ICT in the teaching-learning process. ICT can play a very influential role in the following crucial areas of teaching-learning:

- Communication (between teacher & learner and learner & learner):** Communication or interaction is an essential part of any teaching learning process. Interaction between teacher and learner is the base of our learning process, but peer interaction is also significant for effective learning. While highlighting the role of modern information and communication technology in education, Wedemeyer (1978) stated that there are four essential elements in a teaching-learning situation. They are the teacher, the learner, the communication system or mode, and the content/curriculum (something to be taught/learned). Wedemeyer argued that in earlier days, the speech was the only communication mode between a teacher and a learner, so it was obviously required for teachers and learners to be at the same place and same time for the teaching-learning activities. The time and distance between teacher and learner are dependent upon the communication mode. Nowadays, advanced communication technologies are available, and the speech can easily be replaced with them. The integration of communication technology liberates the education system from the 'space' barrier. Teaching-learning can happen even after the class timings or during vacations or through distance mode. ICTs can provide virtual platforms for the teacher learner interactions and peer interactions as well.
- Content generation:** ICTs have vast potential to create effective educational content for all levels of education from primary classes to higher education. With the capacity to converge various media, it can produce excellent learning materials equipped with text, graphics,

animations, photographs, audio, video etc. Complex concepts can be ICT for Education demonstrated with the help of videos and animations more easily and interestingly. Hypertextuality gives an opportunity to provide more and more in-depth information.

With the help of modern ICT tools, we can create various simulations which can make our learning more useful and interesting. Simulations can provide experiences very near to reality in a cost-effective manner. You can take the example of virtual labs. Virtual labs can provide practical facilities to the students of educational institutions situated in backward areas and struggling for proper infrastructure.

- iii. **Storage and Access to the Content:** When we discuss the scope of ICT in the teaching-learning process, storage of vast amounts of e-content and providing smooth access to these content are the primary concern. ICT has made these things possible at a very minimal cost. E-repositories can be created, and learners can get access from any part of the world.

ICT has completely changed the scenario of knowledge creation and dissemination. A considerable amount of information is available on the Internet on both categories, free of cost and paid. You can access this ocean of knowledge from any corner of the world. Open Educational Resources (OER) are also by product of ICT. These free educational resources are beneficial for learners as well as teachers. It provides the environment of collaborative knowledge creation and consumption across the globe at almost zero cost. Now we shall discuss the scope of ICT in the management of educational institutions which is also an important area.

9.2.2 ICT for Management of Educational Institutions

Apart from core teaching-learning activities, educational institutions are also involved in other management related works. Management of educational institutions encompasses several activities like- admission, examination, record-keeping, communication with parents, result declaration, attendance, scholarships, transportation, leaves, finances, etc. ICT is playing a very crucial role in these management related activities. Various ICT based solutions or systems are available with different modules which can take care of teaching-learning as well as administrative activities of educational institutions from school to university levels.

Government educational institutions, from primary schools to universities, are the only destinations for a significant segment of our population, mostly belonging to the weaker sections. Effective and transparent management of government educational institutions can enhance the quality of education in these public-funded institutions, which finally help to achieve the goal of quality education for all. Here, ICT can play a significant role. For example, an ICT based attendance system for teachers and students can ensure the conduct of proper classes. The central government and many state governments are taking steps to integrate the information and communication technology in the management system of their institutions, but it is not a

smooth path. There are various challenges and hurdles too. We shall discuss these issues and challenges in section 13.6.

Activity- 1

There are many online courses available on the Internet. Many are also free. SWAYAM is one such platform that offers free online courses. Enrol in any short course and try to analyse whether that course has used all the benefits of ICT to enhance the learning experience of students or not.

9.3 ICT IN EDUCATION: MAJOR REQUIREMENTS

No doubt, effective integration of modern ICT in education has many advantages, but it needs congenial frameworks and environments. Proactive leadership, favourable policies and financial support may create such an enabling environment where successful integration can happen. Here, we shall discuss some critical requirements for effective integration of modern ICT in education. These are as follows:

- i. **Proactive Leadership:** Proactive leadership is the key to successful incorporation of any change or any innovation in any field of our life. The same is applicable here also.
- ii. **Favourable Policies:** Enabling policies to prepare the right path for any transformation. Supporting policies may steer all the stakeholders towards adopting ICT enabled education systems. We shall discuss the various policies of the government that support technology integration in the later part of this Unit.
- iii. **Financial Support:** Nothing can be done without financial resources. Adequate financial support is a must for the successful integration of ICT in the Indian education system.
- iv. **ICT Infrastructure:** Reliable ICT infrastructure is a prerequisite for ICT enabled education. It encompasses a variety of things such as – devices (computers, laptops, smartphones, tablets, etc.), software, high speed and affordable Internet services, etc. If we want to enjoy the fruits of ICT enabled education, we need adequate and dependable ICT infrastructure across the country.
- v. **Development of Digital Content:** After having good ICT infrastructure, now we shall use this infrastructure for two essential purposes – development of digital content and dissemination of these content. Development of quality digital content is one of the major requirements for the successful use of ICT in the field of education. With the help of modern ICT tools, excellent digital content equipped with all the advantages of modern technology can be produced. We all know that digital content provides an opportunity to converge all the media elements. Texts, photographs, graphics, animations, audios and videos

and all types of content can be integrated with each other along with the interactivity. This type of multimedia oriented and authored content improves the learning experience of the students.

- vi. **Dissemination and Sharing of Digital Content:** Proper dissemination of the digital content is equally important. In order to distribute digital content among the target population requires a fair amount of financial and human resources. Many times, due to lack of proper dissemination, learners are deprived of quality study materials.
- vii. **Capacity Building of Teachers:** Teachers are one of the main ICT for Education stakeholders of the education system, and without their active participation, any attempt to integrate modern technologies in education will undoubtedly fail. In this context, the capacity building of teachers includes three significant aspects: a. teachers should be aware of the advantages of ICT enabled education; b. they should be skilled enough to integrate modern ICT tools in their teaching practices effectively, and c. they should be trained to create and disseminate the quality digital study material.
- viii. **ICT Literacy among Learners:** Learners are the centre of focus in any educational transaction. They are the end-users. If they do not know how to utilise the ICT enabled learning facilities, the whole process of ICT integration will go in vain. So, ICT literacy of learners across the country is one of the main requirements for successful integration of information and communication technology in the Indian education system.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit.

1. How can ICT support in the creation of quality study material?

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2. Why is ICT literacy of the learners required for the successful integration of modern Information and Communication Technology in Indian educational system?

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9.4 ICT IN EDUCATION: INDIAN SCENARIO

What is your observation about the utilisation level of ICT in education in India? Answers may vary from person to person, but nobody can deny that we are progressing. We can quickly notice that a large section of students, teachers and non-formal learners are using digital technologies for their learning purposes. In this section, we shall discuss the use of ICT in education in the Indian context.

9.4.1 Use of ICT in Education

Being the second most populated country in the world, India has an extensive education system in terms of a number of institutions. The whole education system may broadly be divided into two parts, i.e. school education and higher education. There are 15, 22, 346 schools (primary, upper primary, secondary and senior secondary) and 51,793 all types of higher education institutions running in this country (MHRD, 2018).

In India, both the public (Government) and private sectors are actively involved in the field of education, and both are trying to integrate more and more ICT in education. Private sector looks more active on this front, but government institutions are also trying very hard to provide the benefits of modern digital technologies to their students and teachers. The Indian education world is integrating ICT in both the fields, teaching-learning activities as well as management of the institutions. University Grant Commission (UGC) has instructed all the higher education institutions to adopt the online admission system ultimately. Online examinations activities are quite common. The National Testing Agency (NTA) was established in 2017, and it is conducting massive (in terms of a number of students) examinations through online across the country. Institutions are adopting various educational technologies like smart classrooms, digital learning materials, Learning Management System (LMS), etc. Social media is also being used for educational activities. You may find thousands of YouTube channels providing lectures on different subjects in different languages. Both government and private sectors are taking steps to enhance the level of ICT integration in educational transactions. We shall discuss these steps in the coming sub-sections of this Unit.

9.4.2 ICT in Open and Distance Education

Distance education allows the learner and teacher to be physically apart during the teaching-learning process and maintain communication in a variety of ways (Keegan, 1986). The need for maintaining communication in a variety of ways demands various communication technologies in open and distance education. According to Moore (1989), interactivity is a necessary ingredient of the education process and effectiveness of distance education depends upon the level of interactivity provided to the learners and instructors. For enhancing the level of interactivity, various contemporary technologies may be used.

You can observe that both scholars emphasised the use of modern and advanced communication technology to make distance education more effective. Thus, ICT has a more significant role in open and distance education in both segments: administrative activities and academic transactions. On the basis of the level and type of technological interventions, UNESCO (2002) classified open and distance learning (ODL) into the following three generations:

- i) **The First generation:** The first generation ODL programmes were popularly known as correspondence courses. The print was the only form of teaching-learning content, and postal services were used for delivery of the content and communication between teacher and learner.
- ii) **The Second generation:** This generation came after the establishment of open universities. **Print remained a significant component**, but other media were also introduced. Audio, video and multimedia programmes were produced and sent to the learners along with the print materials. Besides postal services, radio, television, telephone and face-to-face contact sessions were used to communicate with learners.
- iii) **The Third generation:** It has the components of earlier generations, but the two-way communication between teacher and learner has become relatively fast and effective. Internet-based e-learning platforms have ICT for Education converged all the media elements and providing communication tools like audio and video chat. This is the generation of e-learning and mobile learning.

We can easily witness that the majority of the ODL programmes belong to the second generation, but we are trying to move towards the third generation of ODL. Now, you can understand the significant role of ICT in distance education because more ICT integration will improve the generation of our distance programmes.

9.4.3 Government Interventions

In a democratic country like India, where we believe in the concept of the welfare state, the government plays a vital role in the field of education as it is considered an essential factor for inclusive development. As far as the integration of ICT in education is concerned, the government can intervene mainly in two ways. First, by formulating policies which can validate, regulate and boost the use of ICT in education at various levels. If the government formulates favourable policies, they can instruct all the stakeholders, including private educational institutions, to adopt more ICT enabled education. Moreover, second, the government can run their own projects. Through these government-funded projects, the government can promote more ICT enabled education in the country. Here, we shall discuss both types of government interventions.

i. Government Policies on ICT for Education

In India, the government has been emphasising the use of technology in education at all levels for a long time. The National Policy on Education (NPE), 1986 (modified in 1992) has also highlighted the potential of educational technology and the need for technological integration in educational transactions. NPE (1986) says, “Modern communication technologies have the potential to bypass several stages and sequences in the process of development encountered in earlier decades. Both the constraints of time and distance at once become manageable.”

After 1986, the government of India came up with several policies for promoting the use of ICT in education at both the levels – school education and higher education. The National Curriculum Framework (NCF) 2005 has also highlighted the significant role of ICT in school education. The revised National Policy on Information and Communication Technology in School Education came in 2012 as a comprehensive policy, including all aspects of ICT integration in school education in the context of our country. This policy talks about ICT literacy among teachers and students, ICT for the students with special needs, ICT infrastructure, use of ICT in school management, development of digital learning content, dissemination and sharing of digital learning material, ICT for open and distance learning, capacity building of teachers and educational administrators, financial sustainability and monitoring.

The Government feels the need for technology integration in higher education too. They are keen to use various technological resources to fulfil their mission to provide quality higher education to all students (MHRD, 2016). Recent regulations show the government’s emphasis on ICT integration in higher education. We can discuss the few following significant regulations which will promote the use of ICT in higher education:

UGC (Credit Framework for Online Learning Courses through SWAYAM) Regulation, 2016: The Government of India announced its policy on blended learning in higher education, the mixture of conventional classroom teaching and e-learning. The University Grant Commission (UGC) notified the rules and regulation for blended learning. Students of conventional face-to-face mode can earn a part of their programme credits through MOOCs (Massive Open Online Courses) available on SWAYAM (Study Webs of Active Learning for Young Aspiring Minds) platform. This step may be considered as an essential milestone for technology-enabled learning in India.

University Grants Commission (Open and Distance Learning) Regulations, 2017: This regulation also encourages the active integration of ICT in distance learning programmes. It allows the integration of ICT based learning components in every element of a distance programme such as study material, contact classes, assignments and practicals. According to this regulation, institutions offering distance programmes can utilise virtual labs for practical sessions (up to 20%).

University Grants Commission (Online Courses or Programmes) Regulations, 2018: It can be considered a revolutionary regulation which

allows complete online courses or programmes. In these programmes, the highest level of ICT integration can be seen.

National Education Policy 2020: In a major shift, NEP gives a higher importance to technology enabled inclusive learning.

These regulations give clear indications that the government is quite keen to promote ICT in educational transactions at all levels.

ii. Government Projects

The Government of India is of the view that the three fundamental principles of its Education Policy viz., Access, Equity and Quality can be achieved through the use of ICT in education. The National Mission on Education through Information and Communication Technology (NMEICT) is the result of it (MHRD, 2016). NMEICT is an umbrella project for all the initiatives in the field of technology integration in Indian higher education. The government is working on two significant fronts. First, the creation of quality digital learning material in all disciplines and second, creation of networks and platforms to share the knowledge among all higher education institutions across the country. SAKSHAT has been developed as a free one-stop online delivery platform for all the generated digital content under NMEICT. Programmes such as the National Program on Technology Enabled Learning (NPTEL), Virtual labs, Talk to a Teacher, National Digital Library, Vidwan (Expert Database and National Researcher's Network), E-Yantra, E-Kalpa, Study Webs of Active Learning for Young Aspiring Minds (SWAYAM) and SWAYAM PRABHA are running under the umbrella of NMEICT. SWAYAM has been developed to offer online courses, whereas SWAYAM PRABHA is a group of 32 DTH educational channels for different academic disciplines.

Though the primary focus of SWAYAM is on higher education, school level ICT for Education online courses are also available on this platform. SWAYAM PRABHA is also with similar options. While a majority of the channels are dedicated to higher education, yet a few channels are devoted to school-level education too. ePathshala, eBasta, ICT @ Schools Scheme, etc. are a few other essential government projects at the school level.

9.4.4 ICT Enabled Education and Private Sector

The private sector too has a strong presence in Indian education. They are involved in all types of education, for example - formal education, tutoring, preparation of competitive examinations, etc. At the same time, they are also the big promoters of ICT enabled education in India due to various reasons. If we talk about formal education, the private sector is increasing technological integration at both the levels, school education as well as higher education. Private schools, colleges and universities are adopting various educational technologies like smart classrooms, digital learning materials, virtual labs, (LMS), etc.

You can quickly notice that ICT is being used in tutoring and preparation for competitive examinations in a big way. A number of advertisements of

various learning apps and online coaching can be seen on all media platforms. The private sector is utilising the capabilities of modern ICT tools to provide quality study materials in a more interesting and easy to understand the way to their target population. Market competition compels them to be more innovative and creative. They are using various components such as-video lessons, tests, quizzes, games, interactive lessons, live streaming, etc. They are also merging educational components with virtual reality which gives a unique experience to the students and also help them to get clarity on various concepts that need more practical explanations. Apart from these, they are providing very detailed performance analysis to the students. You must have used or noticed many educational apps for tutoring and preparation of various competitive examinations. Gradeup, Unacademy, Toppr, Adda247, Vedantu, STEPapp, BYJU's, etc. are a few examples.

Apart from formal education and coaching, the private sector is actively involved in creating platforms for online courses. Nowadays, the needs of various job providing sectors are changing very fast. Employees need to be updated with fast-changing technologies and knowledge. Micro degrees that are highly focused on specific skills are in huge demand, and many private players have developed different platforms to offer various online courses relevant to the employees and job seekers. Coursera, edX, Udacity, FutureLearn, Udemy, etc. are a few examples of online course platforms run by the private sector.

Activity- 2

Interview any five students who have completed or pursuing any type of online course. As per their experience, compare the online learning environment with the conventional classroom learning method.

Check Your Progress: 2

Note: - 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit.

1. What is NMEICT?
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2. What is the third generation open and distance learning?
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3. SWAYAM stands for
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 - a. Study Webs of Active Learning for Young Aspiring Minds
 - b. Study World of Active Learning for Young Aspiring Minds
 - c. Study World of Advanced Learning for Young Aspiring Minds
 - d. Study Webs of Active Learning for Youth Advanced Minds
4. SWAYAM PRABHA is a/an
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 - a. Online learning platform
 - b. Institution running ICT enabled programmes
 - c. Group of 32 DTH educational channels
 - d. Internet service provider

9.5 INTEGRATION OF ICT IN EDUCATION: ISSUES AND CHALLENGES

We all agree that ICT has tremendous potential to bring about positive transformation in all dimensions of Indian education. It can help us to achieve the goal of providing quality education to all. As a country, we are working hard to utilise the advantages of modern ICT tools for our education system, and we have achieved many successes too. However, we have still to cover a long distance. India is full of diversities, and the picture of rural and remote tribal areas may be completely different from the cities in this context. Even in the cities, there may be different levels of ICT integration in government educational institutions and private ones. You may observe that in the same city, there are few private schools having world-class ICT integration in their educational transactions and on the other hand, many government schools are still struggling for the basic ones. So, the successful integration of ICT in education is not an easy task in a developing country like India. There are some hurdles also. Few hurdles are universal and common in all cases, but few are country-specific due to the social and economic status of that particular country. Now, we shall discuss the issues and challenges in the effective integration of ICT in the Indian education system. These are as follows:

- i. **The Attitude of Teachers and Students:** Users' attitude is one of the universal challenges for the effective integration of ICT in educational transactions. Emphasising the importance of teachers' ICT for Education attitude, Johnson (2012) states that the attitude of teachers towards educational technology is one of the crucial factors influencing the process and success of technology integration in higher education. Teachers should have complete clarity on why technology is important for education and why they should change their role from 'sage on the stage' to 'guide on the side'. Apart from teachers, students should also have a positive attitude towards ICT enabled learning. In brief, perceived usefulness and perceived ease of use for ICT enabled teaching-learning should be on a higher side among students and teachers.
- ii. **Digital Inequality:** You must have learnt the term 'Digital Inequality in Unit 3 in this course. Digital inequality refers to the differences in access and use of digital technology between two or more groups of people. There will be a digital inequality if one group of people has higher access to ICT in comparison to another group. In the case of digital inequality, the digitally marginalised section is deprived of the benefits of modern information and communication technologies. It is one of the significant challenges for the successful integration of ICT in the Indian education system because India is suffering from an extensive digital inequality scenario.

Digital inequality may be existing due to several reasons like - education, ICT literacy, socio-economic condition, language, area of residence (rural, urban, remote tribal), age, etc. We all know that India is a country full of diversity. We have a diverse population on the basis of language,

residential area, income, education and socioeconomic conditions. We still have a large section of the population illiterate. Take the example of a student from a small town of Uttar Pradesh. She has proper access to ICT, and she is skilled too, but her medium of education is Hindi. She wants to use ICT to get better subject contents, but most of the content available on the Internet is in English. Thus, despite having access and skill, she suffers due to digital inequality. Here, language is the cause of digital inequality. Similarly, many Indian languages are facing this crisis.

- iii. **Availability of Digital Content in Indian Languages:** India is a multi-linguistic country having 22 official languages listed in the 8th schedule of its constitution. We all know that an extensive section of our population is getting an education in their own mother tongues, for example – Hindi, Bengali, Tamil, Telugu, Marathi, Gujrati, Malayalam, etc. Thousands of schools, colleges and universities are providing education at various levels in Indian languages. We have already discussed an example where language caused a digital inequality. Therefore, if we want to extend the benefits of modern ICT to all the students without any discrimination, creation of digital study materials in Indian languages is a must, and at the same time, it is a big challenge too.
- iv. **Infrastructure:** For effective integration of ICT in Indian education, we need reliable information and communication infrastructure across the country, and it is a big challenge. We all know that the availability of adequate ICT infrastructure creates a digital inequality between urban and rural areas (we have covered the access and penetration of ICT in Indian in Unit 3 of this course). We need to equip our rural and remote tribal areas with high-speed, dependable Internet services only then can we achieve our dream goal of transforming India into a digitally empowered society and knowledge economy. This is also the vision of “Digital India” programme.

Reducing the digital inequality between urban and rural areas has been on the government’s priority list. In 2011, the government of India had approved the setting up of National Optical Fibre Network (NOFN). Project Bharatnet has been started to connect 2,50,000 Gram Panchayats with broadband. Bharat Broadband Network Limited (BBNL), a government of India undertaking, is continuously working on it. As on 27th December 2019, optical fibre cable of 3,99,375 Kms of length has been laid down in 1,44,670 Gram Panchayats (BBNL, 2019). The efforts show that providing ICT infrastructure to all in a non-discriminatory manner is a priority, and we shall overcome the challenge of infrastructure one day.

- v. **Capacity Enhancement of Related Human Resources:** Effective integration of technology in education needs skilled stakeholders. Capacity enhancement of students, teachers and other support staff is a need and at the same time a challenge also. The gap between ICT skills of different groups of teachers and students also cause digital inequality and put a hurdle in the way of smooth and inclusive integration of ICT in education without any discrimination. If you compare, you will find a

massive gap between the ICT capabilities of teachers coming from metros and remote rural areas. It is a big challenge, and we have to enhance the ICT related capacity of our teachers, students and support staff coming especially from rural areas.

- vi. **Other Issues:** Apart from the above-mentioned challenges, there are also some other issues which need to be discussed here. For example, - the quality of Open Educational Resources (OER). OER provides learning content free of cost to the learners, but the quality of OER may be a valid concern. However, with the active participation and cooperation of the teaching fraternity can minimise this problem. ICT enabled education puts our small and vulnerable kids online, and in this case, E-safety of online children is also an important issue.

Activity- 3

Search the various platforms offering online courses, including SWAYAM. Find out the courses available in Indian languages and then write a note on 'Online Courses in Indian Languages'.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit.

1. Explain the term 'Digital Inequality?'

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2. What are the significant challenges in the successful integration of ICT in the Indian education system?

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9.6 LET US SUM UP

Technology is definitely impacting human life. Advancements in Information and Communication Technology (ICT) affect each and every field of human

civilisation, including education. In this Unit, we have discussed the different aspects of the integration of ICT in education. We started with the scope of ICT in education and then talked about significant requirements for the successful integration of ICT in it. The status of ICT enabled education in India, and the roles of public and private sectors were also discussed. We all know that India is a developing country with a considerable population and many kinds of diversities. Here, ICT has emerged as an effective solution to provide quality education to this huge population in a cost-effective manner, but there are some issues and challenges too. We have discussed this aspect also in this Unit.

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9.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. ICT has vast potential to create effective educational content for all levels of education from primary classes to higher education. With the capacity to converge various media, it can produce excellent learning material equipped with text, graphics, animations, photographs, audio, video etc. Complex concepts can be demonstrated with the help of videos and animations more easily and interestingly. Hypertextuality gives the opportunity to provide more and more in-depth information. With the help of modern information and communication technology, we

can create various simulations which can make our learning more useful and interesting. Simulations can provide experiences very near to reality in a cost-effective manner.

2. Learners are the centre of focus in any educational transaction. They are the end-users. If they do not know how to utilise the ICT enabled learning facilities, the whole process of ICT integration will go in vain. So, ICT literacy of learners across the country is one of the main requirements for successful integration of information and communication technology in the Indian education system.

Check Your Progress: 2

1. The Government of India thinks that the three fundamental principles of its Education Policy viz., Access, Equity and Quality could be achieved through the use of Information and Communication Technology (ICT) in education. The National Mission on Education through Information and Communication Technology (NMEICT) is the result of it. NMEICT is an umbrella project for all the initiatives in the field of technology integration in Indian higher education.
2. Third generation open and distance learning has the components of earlier two generations, but the two-way communication between teacher and learner has become relatively fast and effective. Internetbased e-learning platforms converged all the mediums and provided communication tools like audio and video chat. This is the generation of e-learning and mobile learning.
3. a
4. c

Check Your Progress: 3

1. Digital inequality refers to the differences in access and use of modern information and communication technology (digital technology) between two or more groups of people. There will be a digital inequality if one group of people has higher access to ICT in comparison to another group. In the case of digital inequality, the digitally marginalised section deprives the benefits of modern information and communication technologies.
2. The significant challenges in the successful integration of ICT in the Indian education system are digital inequality, the attitude of teachers and students, ICT infrastructure, capacity enhancement of related human resources and content creation in Indian languages.

UNIT 10 ICT FOR HEALTH

Structure

- 10.0 Introduction
- 10.1 Learning Outcomes
- 10.2 Health Sector and ICT
 - 10.2.1 Need for Health Communication
 - 10.2.2 ICT in Health Communication
 - 10.2.3 Health IT
- 10.3 Health Information Online
 - 10.3.1 E-Health
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10.0 INTRODUCTION

In this Unit, we shall discuss the role of ICT in communicating health messages, how to seek health information from the Internet, strategies and skill development for acquiring health information and significant barriers and problems in acquiring health information.

Health communication focuses on the development of effective messages through different media. It serves as a critical platform for health literacy

messaging. Even though the term health communication is used in a broader sense, this has a long history.

Ensuring healthy lives and promoting well-being at all ages is essential for social development, particularly as part of the Sustainable Development Goal (SDG 3). Significant progress has been made in improving the health of millions of people. However, at least half of the world's population is still without access to essential health services and information.

For promoting and protecting health, of yourself and others, you can raise awareness in your community about the importance of good health, healthy lifestyles as well as people's right to quality health care services, especially for the most vulnerable such as women and children. You can also hold your government, local leaders and other decision-makers accountable to their commitments to improve people's access to health and health care. However, for all these to work out, you should have some knowledge about what it is, what to communicate through which medium, how to collect information and so on. In this information age, technology has a significant role in communicating information. Through better usage and improvement can ensure health lives.

In this Unit, we shall discuss the growing significance of Information and Communication Technology (ICT) in the fields of public health and community welfare.

10.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- understand the dynamics of health communication;
- explore the role of ICT in health communication;
- explain about Health Information Literacy in the Internet age;
- understand skill acquisition-theory and models; and
- explain the information dissemination-barriers, problems and issues.

10.2 HEALTH SECTOR AND ICT

Health Communication is the study and use of communication strategies to inform and influence decisions and actions to improve health. It has clearcut objectives to promote changes, create awareness, knowledge, and if needed, appropriate changes in attitudes and beliefs.

Strategic approaches for communicating health messages and information have been practised informally for years. The term health communication was first used in 1961 when the National Health Council organised a National Health Forum to discuss challenges faced in the communication of health information (Helen, 1962). The term was used again in 1962 when Surgeon General Luther Terry organised a conference on health communication to discuss how various techniques of communication can be used to make health information available to the public (US Department of

Health Education and Welfare, 1963). The term was adopted officially by members of an interest group at the International Communication Association ICA in 1975. And, in 1997, the American Public Health Association recognised health communication as a discipline within the broader fields of Public Health Education with health promotion and health communication as subjects to undergo comprehensive growth and change.

10.2.1 Need for Health Communication

Before understanding the need for health communication, we should know the six principles given by the World Health Organisation which describes the role of communication system in the health sector.

1. **Accessible to decisions-makers:** Communicators should identify all channels that are available and map their comparative capacities to reach priority audiences. Using the right mix of channels helps empower audiences with the information they need in order to make informed decisions.
2. **Actionable by decisions-makers:** Communicators must understand the knowledge, attitudes and behaviours of the target audience in order to create messages that address barriers and encourage decisionmakers to take the recommended steps.
3. **Credible and trusted as perceived by decision-makers:** The reputation of WHO is a crucial factor in deciding whether individuals will take action on its advice and guidance. The more decision-makers trust WHO, the more likely they will believe, and act on the information communicated by the Organisation.
4. **Relevant to decisions-makers:** In order to be relevant, communications must help decision-makers understand that there is a health risk for themselves or for those they care about, that the risk could be severe and that they can reduce that risk by undertaking recommended actions.
5. **Timely to enable decision-making:** Across all health issues, WHO must make information, advice and guidance available in a timely way, so audiences have the information they need when they need it in order to make appropriate health decisions.
6. **Understandable to decision-makers:** The WHO communicates with a wide variety of decision-makers. Some have backgrounds in medicine and public health and rely on WHO for technical information. However, many key audiences are not technical experts. They need information that is easy to understand in order to understand health risks and take appropriate actions. From the above perspective there arises the need for health communication which can reduce the level of awareness by:
 - Increasing risk perception;
 - Reinforcing positive behaviours;
 - Influencing social norms;
 - Increasing the availability of support and needed services;

- Empowering individuals to change or improve their health conditions.

A range of organisational and institutional developments associated with the technological changes have altered not only patterns of production and distribution of goods, but also, even more significantly, patterns of services especially the nature of the delivery of many essential services such as health.

10.2.2 ICT in Health Communication

ICT can play a critical role in improving health care for individuals and communities by providing new and more efficient ways of accessing, communicating, and storing information. ICTs can help bridge the information divide that has emerged in the health sector in developing countries—between health professionals and the communities they serve and between those involved in health research and the practitioners who need it through the development of databases and other applications. ICTs can also provide the capacity to improve health system efficiencies and prevent medical errors.

ICT can affect health conditions in developing countries both directly and indirectly. It can work directly by improving health care provision and disease prevention. It also works indirectly on the health status of the population through its effects on the broader determinants of health, such as growth, the economic position of households, and the social infrastructure.

The first is its role as an instrument for continuing education and lifelong learning that will enable doctors in developing countries to be informed about and trained in the use of advances in knowledge.

The second is its use as a delivery mechanism to poor and remote locations of a wide variety of services varying from improved public health education to emergency advice, including advice on dealing with and mitigating the consequences of natural disasters.

The third is its potential use as a mechanism to increase the transparency and efficiency of governance, which would, in turn, improve the availability and delivery of publicly provided health services.

10.2.3 Health IT

The term “Health Information Technology” (Health IT) refers to the electronic systems health care professionals, and increasingly, patients use to store, share, and analyse health information. Technological innovations have greatly improved the ability to access information, facilitate communication and improve efficiency in nearly all aspects of life. Similarly, the use of technology to directly impact the health of populations, as well as to provide access to crucial public health information, holds great promise that is emerging at the forefront of medical care and public health industries. Technology can also cut across the barriers to reaching out to diverse populations to meet their needs, and using health information technology interventions for vaccinations.

Check Your Progress: 1

Note: - 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. Explain the need for ICT in Health.

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2. Mentions various applications of e-health.

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10.3 HEALTH INFORMATION ONLINE

You may be turning to the Internet to obtain information about health and health care. Internet use has grown dramatically over the last decade, and searching for health information is an essential application for many users. What makes you search the Internet for health information? Have you ever been confused by seeing lots of information? You may be confused the complexity, which is the right one to choose? With the emergence of internet health information, its broader aspect has been growing, and new branches have emerged in the health sector.

10.3.1 E-health

There are many terms used to describe the use of ICT in supporting healthcare.

These terms are often used interchangeably, and thus can contribute to misunderstanding or misconceptions. Terms include eHealth, mHealth, tele-health, tele-rehabilitation and telemedicine. For instance, the World Health Organization (WHO) uses the term e-Health to describe the use of ICT to support public health and healthcare areas. E-health is an emerging field in the intersection of medical informatics, public health and business, particularly its application in health services and information delivered or enhanced through the Internet and related technologies. In a broader sense, the term e-health characterises not only a technical development, but also a state-of-mind, a way of thinking, an attitude, and a commitment for

networked global thinking to improve health care locally, regionally, and worldwide by using ICT.

This definition hopefully is broad enough to apply to a dynamic environment such as the Internet and at the same time, acknowledges that e-health encompasses more than just “Internet and Medicine.”

As such, the “e” in e-health does not only stand for “electronic,” but implies several other “e’s,” which together best characterises what e-health is all about (or what it should be).

Efficiency - one of the promises of e-health is to increase efficiency in healthcare, thereby decreasing costs.

1. **Enhancing quality of care** - increasing efficiency involves not only reducing costs but at the same time improving quality. E-health may enhance the quality of healthcare, for example, by allowing comparisons between different providers, involving consumers as additional power for quality assurance, and directing patient streams to the best quality providers.
2. **Evidence-based - e-health** interventions should be evidence-based in the sense that their effectiveness and efficiency should not be assumed but proven by rigorous scientific evaluation. Much work still has to be done in this area.
3. **Empowerment** of consumers and patients. By making the knowledge bases of medicine and personal electronic records accessible to consumers over the Internet, e-health opens new avenues for patientcentred medicine and enables evidence-based patient choice.
4. **Encouragement** of a new relationship between the patient and health professional, towards a true partnership, where decisions are made in a shared manner.
5. **Education** of physicians through online sources (continuing medical education) and consumers (health education, tailored preventive information for consumers).
6. **Enabling** information exchange and communication in a standardised way between healthcare establishments.
7. **Extending** the scope of health care beyond its conventional boundaries. This is meant in both a geographical sense as well as in a conceptual sense. E-health enables consumers to obtain health services online from global providers quickly. These services can range from simple advice to more complex interventions or products such as pharmaceuticals.
8. **Ethics** - e-health involves new forms of patient-physician interaction and poses new challenges and threats to ethical issues such as online professional practice, informed consent, privacy and equity issues.
9. **Equity** - to make healthcare more equitable is one of the promises of e-health, but at the same time there is a considerable threat that e-health

may deepen the gap between the “haves” and “have-nots.” People who do not have the money, skills, and access to computers and networks cannot use computers effectively. As a result, these patient populations (which would benefit the most from health information) are those who are the least likely to benefit from advances in information technology, unless political measures ensure equitable access for all. The digital divide currently runs between rural vs urban populations, rich vs poor, young vs old, male vs female and between neglected/rare vs common diseases.

In addition to these essential e’s, e-health should also be

- easy-to-use,
- entertaining (no-one will use something boring!) and
- exciting- and it should exist.

ICT for health (or e-health) programs are often considered to be expensive, time-consuming, risky, and otherwise distracting from the primary focus and intent of health sector programs. In some cases, these criticisms may be valid. There are, however, a wide range of low-cost and sustainable ICT for health program components that can augment capacity and improve the overall effectiveness of health development programs. e-Health literacy is a concept that combines health literacy and also information literacy, along with an addition of traditional literacy, media literacy, computer literacy and scientific literacy.

10.3.2 E-Learning

E-learning can also be termed as a network-enabled transfer of skills and knowledge, and the delivery of education is made to a large number of recipients at the same or different times.

With rapid progress in technology and the advancement in learning systems, it is now embraced by the masses. The introduction of computers was the basis of this revolution and with time, as we have got hooked to smartphones, tablets, etc. Learning has a broad scope in the health sector. It aims to improve the coverage and quality of health services for rural and remote areas. Through e-learning usage of website portals, audio, video and digital presentations, communications are given to healthcare professionals and workers.

In the health sector, where knowledge and skills require frequent updates alongside growing complex non-communicable diseases and frequent changes in medical techniques and technologies, e-learning platforms support the health workers to effectively implement prevention, management of treatment for various health issues. The e-learning platform which transcends the physical border and locations is also expected to provide a useful forum and venue to frequently facilitate and host a continued technical discussion and peer learning with skilful medical practitioners globally.

10.3.3 Case studies

To take up Digital India initiative ahead, the Ministry of Health and Family Welfare has started various e-Gov initiatives in healthcare sectors in India, and the division is named as eHealth division. e-Health is broadly defined as the use of Information and Communication Technology (ICT) in health. It will make a world of difference in India, where mobile technologies have been penetrating at a rapid rate. As India has a strong presence in IT, the integrated health information system seems to have the potential to serve the needs of all stakeholders, by contributing approximately eight per cent to the GDP.

The eHealth initiatives have the vision to deliver better health outcomes in terms of

- access;
- quality;
- affordability;
- lowering of disease burden and
- efficient monitoring of health entitlements to citizens.

The scope of these initiatives is to make all medical facilities available every time from any part of the world through web services, mobile services, SMS, or Call centre services. Broadly, the intent is to cover online medical consultation, online medical records, online medicine supply management and pan-India exchange for patient information but not limited to the said services.

10.4 STRATEGIES FOR HEALTH COMMUNICATION

The materials reflect the situations that ‘old’ and increasingly ‘new’ technologies are often included in an analysis of media and health communications, as they constitute a large portion of media (only print is not an ‘ICT’, but this definition is also not always apparent, as the literature show that increasingly the Internet is also delivering content that was traditional ‘print’, and there is a convergence between newsprint and internet versions of newspapers and other publications). The gathered literature point to the preeminence of communication in much of the ICT for development discourse, and the importance of the processes of communication in speaking about the effectiveness of ICT usage.

10.4.1 Health Literacy

The concept of health literacy, typically used within health sciences and health communication, is often defined as ‘the degree to which individuals have the capacity to obtain, process, and understand basic health information and services needed to make appropriate health decisions’, the focus is on the individual’s abilities to make decisions. However, there is also a definition

where the cognitive and social abilities, and the aim to promote and maintain an individual's health, are emphasised.

The Health Promotion Glossary of the World Health Organisation defines the concept in the following way: "Health literacy represents the cognitive and social skills which determine the motivation and ability of individuals to gain access, to understand and use information in ways which promote and maintain good health."

10.4.2 Health Information Literacy

As a concept, information literacy was introduced in the early 1970s and has since spread from being used within information management in organisational contexts and also to be shared in other information environments. The approach is increasingly moving away from being focused on individual skills towards attaining individuality.

A related concept that describes health-related information behaviour, includes seeking and using of information related to health or medicine is Health Information Literacy (HIL).

The Medical Library Association defines HIL as the ability to recognise a need for health information, to know how and where to find health information, and how to evaluate and use this information in everyday life to make right health decisions.

The concept is a combination of health literacy and information literacy. An information-literate person has, according to a widely used definition, the ability to determine the extent of information needed; access the needed information effectively and efficiently; evaluate information and its sources critically; incorporate selected information into one's knowledge base; use information effectively to accomplish a specific purpose; and, to understand the economic, legal and social issues surrounding the use of information, and to access and use information ethically and legally.

10.4.3 Information Dissemination

The study and use of communication strategies to inform and influence individual and community decisions that affect health links the fields of communication and health and is increasingly recognised as a necessary element of efforts to improve personal and public health. Dissemination is the targeted distribution of information and intervention materials to specific public health or clinical practise audience. The intent is to spread knowledge and associated evidence-based interventions. Dissemination Health and ICT occurs through a variety of channels, social contexts, and settings.

10.4.4 Perspective of Health Information

Government organisations, media, civil society, community and political leaders and even the academic and research organisations can impart health literacy. The government can play a decisive leadership role in developing and implementing health-related strategies. Additional stakeholders include

the insurance industry, policymakers, and educational training institutions”. The long-term solution lies in “improving the literacy of the population” at large through schools and other communities.

Governments can take up a vital leadership role in developing and implementing health literacy promotion policies by providing sustained funding, setting up special projects, coordinating action across sectors, and conducting health literacy surveillance regularly. Improving and measuring health literacy (both strengths and needs) is particularly important in poorer areas, so that vulnerable populations are empowered to engage in early and sustained health-promoting actions, whether to prevent acute and chronic conditions or to promote active and curative treatments.

Government agencies and institutions, advocacy groups, media organisations, researchers, and other interested stakeholders can all carry out communication activities. They may use various techniques to communicate evidence so that target audiences can understand it better.

Individuals: The individual is the most fundamental target for health-related change since it is individual behaviour that affects health status. Communication can affect an individual’s awareness, knowledge, attitudes, self-efficacy, and skills for behaviour change. For example, counselling on sexual risk reduction as well as HIV testing is an effective way to change behaviour and protect health.

Print, Television, Radio, New Media Campaigns

Inquiry into health communication also involves the examination of many communication channels face-to-face communication between providers and consumers, members of healthcare teams, and support group members etc.

You should have seen and listened to Public Service Advertisement (PSA) in newspapers, magazines, radio and television channels on vaccination, child and maternity care and various other health-related issues. Moreover, in rural areas still, public announcements and awareness programs are being carried out.

A broad range of both personal communication media (e.g., telephone, mail, fax, e-mail) and mass communication media (e.g., radio, television, film, billboards) have also been the focus of much research.

On 7th April 2012, the world’s most extensive communication programme was launched in India. The “Swasth Bharat” programme is being telecast and broadcast through 30 Regional Kendras of Doordarshan and 29 Stations of All India Radio covering 27 States. The half an hour programme was telecast for five days a week in Regional languages at prime time. The objective of the programme was to empower citizens with Information, Education and Communication on health-related issues. The Ministry of Health and Family Welfare is the first Ministry in Government of India to have signed a Memorandum of Understanding with Prasar Bharti to start such an outreach programme to be able to reach out to a broad spectrum of the population through mass media.

Even as more new communication technologies are being developed, and they have been examined as important media for health communication. These new media, especially the interactive computer technologies and the Internet, have become increasingly important sources for relevant health information and support for many consumers and healthcare providers.

Intermediates-Health Workers-ASHA, ANM, Anganwadi Workers

Community Health Workers (CHW) are the link between the government and the common people. CHW educates the population to attain health literacy.

The World Health Organisation (WHO) defines Community Health Workers as “members of communities where they work, selected by and answerable to the communities for their activities, supported by the health system but not necessarily a part of the organisation, and have shorter training than professional workers.” Effective use of communication and technology by healthcare professionals can bring about change in public centred health information and service. There is evidence through experience that their approach to health is comprehensive. Their tasks vary from health service navigation, primary health care and social services provision to community development and advocacy concerning the social determinants of health.

Community Health Workers in India

CHW's programme has been implemented in India from 1950 onwards. However, it was widely accepted and marked in its implementation as the Integrated Child Development Scheme (ICDS) programme of 1975, which was initiated with the support of UNICEF and World Bank. Two local women each were selected and trained to meet the antenatal and postnatal care needs in villages. They were named as Anganwadi Worker and Helper who worked solely in their village and focused on providing food supplements to young children, adolescent girls and lactating women.

The Auxiliary Nurse-Midwife (ANM), was based at a Sub-Centre and visited villages in addition to providing care at the sub-centre.

In 1977, the Swasthya Rakshak Scheme was initiated for villages with male Community Health workers. However, the scheme existed only in some parts, and later the government began to give more importance to maternal and ante-natal care, and more females CHW's were recruited.

In the year 2006, India implemented ASHA (Accredited Social Health Activist) as one of the critical components of the National Rural Health Mission. The main agenda was to provide every village in the country with a trained female community health activist. Selected from the village itself and accountable to it, the ASHA workers is trained to work as an interface between the community and the public health system. ASHA workers stimulated or motivated change in the individual's perception of behaviour or beliefs. ASHA workers focused on the promotion of MCH, including immunisations and institution-based deliveries, for which they received a Health and ICT performance-related fee.

The Ministry of Women and Child Development (WCD) is undertaking a complete revamp of the Integrated Child Development Services, including digitisation of 13 lakh Anganwadis for real-time monitoring of every child and lactating mother. As part of the digitisation drive, 13 lakh Anganwadi workers were to be given smartphones, and their supervisors were to be given tablets. Both the hardware and software for the project was to be provided by the Bill and Melinda Gates Foundation. A memorandum of cooperation between the ministry and the foundation in this regard was signed in May 2019.

The respective state governments were to provide training to Anganwadi workers to help them switch to the new IT-based system. Though not wholly, ASHA workers were also digitally empowered to some extent. Mobile phones as media are now comparable to popular media like radio and television, which have been used for spreading healthcare awareness messages. It has the capability to provide individually tailored and customised services to underserved people. Health could be used by ASHA workers, especially for point-of-care services, provision of health information, counselling, drug adherence, data collection and monitoring. Mobile apps and digital video production training are given to them. Moreover, in the coming years, India may become the largest public sector with digitised community health workers in the world.

Check Your Progress: 2

Note: - 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. Differentiate between Health Literacy and Health Information Literacy.

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2. Explain the role of different media in imparting health information.

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10.5 SKILL ACQUISITION IN HEALTH - THEORY AND MODELS

Skill acquisition refers to how individuals gather and use health information for their health. Whether print or Internet, it provides access to a vast number of documents on health-related topics, but the nature of choosing completely depends upon the skill of users. These skills depend upon an individual's behaviour which contributes to the development of health theories and models.

10.5.1 Health Belief Model

The Health Belief Model (HBM) is one of the models which identifies behavioural health change. The Health Belief Model focuses on individual beliefs about health conditions, which in turn can predict the individual health related behaviour. It is the most widely accepted model for understanding health behaviour. HBM explains the impact of an individual's perception and attitude toward disease and how those perceptions and attitudes impact their health-related decision-making. The model has been evolved as a concern in public health. The Health Belief Model can be used to design short- and long-term interventions. Health belief models can be used to identify key determinants that influence health behaviour.

The essential components of the HBM depend primarily on two variables

(1) the value of a goal to an individual, and (2) the individual's appraisal of the likelihood of action in achieving that goal. It emphasises the role of moderating factors (demographic, social, and structural factors) and individual perceptions (perceived sensitivity, perceived severity, perceived benefits, perceived barriers, cues to action, and self-efficacy) in determining the likelihood of performing a behaviour. HBM is used as a framework to identify social determinants.

Theoretical Constructs

Perceived behaviour and barrier are the reason for individual behaviour. Since HBM is a social behaviour theory, it is used as a conceptual framework for the identification of social determinants of health.

Perceived Susceptibility

In HBM, perceived seriousness and perceived threat are the two factors which can be studied to know how much the individuals are literate. Even though health information and communication is increasing, social determinants act as a barrier to attaining health literacy.

Perceived Severity

The level of health information literacy of an individual can be measured by assessing their knowledge on sharing particular conditions.

Cues of Action-Perceived barriers and perceived susceptibility leads to action, and this action is measured to know about the determinants of health literacy.

Self-Efficacy-In order to meet the communication needs, the best approach to health information should be to transform efficiently, identifying the worth of health information.

Individuals with low perceived susceptibility may deny that they are at risk of contracting a particular illness. The model does not take into account how emotions such as anxiety or fear affect rational thought and decision-making. Even though HBM is the most accepted model for health communication research, the model lacks the measure of the effectiveness of future consequences, self-identity, concern for appearance and perceived importance.

10.5.2 Anderson's ACT theory

Adaptive Concept of Thought is a cognitive theory developed by John R Anderson, an American psychologist. Cognition is the process of understanding or acquiring knowledge and information through thought, experiences and other senses. The architecture is made with the framework or model which shows the process of understanding and acquiring knowledge and information.

This theory tells us several steps and processes to consider every task Health and ICT performed by human beings. It provides a model or framework for the processes that happen in our mind. The framework or model may look like a programming language. The theory works on psychological assumptions of human nature which were derived from numerous scientific experiences and experiments. The main components of ACT are Modules, Buffers and Pattern Matcher.

Every human task can be made into a model by this theory. The models of every person will be different from one another. It incorporates research on automaticity and explains the development of cognitive skills necessary for processing health information from a variety of electronic sources.

10.5.3 Transtheoretical Model

The Transtheoretical Model (TTM) (or Stages of Change Model) is a cyclic model developed by Prochaska and Diclemente (1983).

The model suggests that people change their behaviour at certain stages in life, rather than making one significant change. During these incremental stages, they consider whether or not to make changes to their behaviour. This cyclic model is based on the premise that people at different levels of readiness change, and during the change process, they move through a series of stages. People move from pre-contemplation (not ready to change) to contemplation (thinking of change), to preparation (getting ready to change), to action (performing the change), to maintenance (continuing the change), to relapse (abandoning changes and reverting to former behaviours). A person

may start at any of these stages and may move between stages. This theory finds application in acquiring health information of individuals as they are changing behaviour at specific points of life and can add their information gathering habits.

These are only some of the theories and models which find application in health information and the Internet, as online health care is enabling the growth of the practitioner-patient relationship and as the new roles of health workers and caretakers are refining, all these models and theories are also changing.

10.6 BARRIERS TO HEALTH INFORMATION LITERACY

Socio-cultural disparity and superstitious beliefs exist in most sections of society. Access to healthcare is still in a deprived condition due to the cultural backwardness and the belief system prevailing within them. They are linked with culture, language and facets of life that vary widely among socio-economic conditions. People with low levels of education and occupation is also a factor. Those who are using and depending on technology for their information seem to be more interested in HIL. Language and communication issues also form barriers. Media literacy forms the basis for this. Vulnerable people face problems in communicating with health professionals and depend more on traditional healthcare providers prevalent in their area. In the global world, migration and immigration are increasing, and hence, the language may form a major barrier.

10.6.1 Credibility of Online Information

The Internet provides you access to a vast number of documents on health related topics. You may be getting many health tips on your mobile phone, have you ever checked the credibility of it? Mobile phones are the most supportive in communicating health literacy and information. It is also a cost-effective medium for message transmission and information. To become savvy information consumers, young people may need assistance in understanding the various health media, in order to build awareness and their own skill sets and to improve their ability to make evidence-based decisions. However, the credibility of websites and trust worthiness in information is an issue.

10.6.2 Problems of Usage

The competence and skill of the people can only assure better health in society. More motivated people, i.e. interested in health information and more active at obtaining it, are also more confident in their ability to do so. Lack of knowledge or confidence to seek health information is a barrier to obtaining such information. Information and reports are increasing day-by-day, but the authenticity of the information and quality of information is still in debate. Evidence-based practices and public awareness are the best way to ascertain it. Government initiatives towards training and training to

access the information to health workers and caretakers are essential in the information loaded world.

10.6.3 Social Determinants

According to the World Health Organisation (2008), health disparities are associated with gradients in social determinants such as income, age, education, and marginalisation or disadvantages due to race and gender.

Educational level, age, gender, cultural beliefs, low socio-economic condition, language and communication strategy all form the social determinants. Family culture, ability and interest to learn all form the determinants of health literacy. Safe housing, transportation, and neighbourhoods, racism, discrimination, and violence, education, job opportunities, and income, access to nutritious foods and physical activity opportunities, polluted air and water, language and literacy skills all determine the individual's health.

10.6.4 Issues in India

India is one of the leading countries which is trying to provide quality healthcare services as it spends 5.2% of the GDP every year. The primary healthcare centres are the main source of healthcare service in rural areas other than secondary care and tertiary care services provided in urban areas where information technology is used, especially in urban hospitals. Specific characteristics such as cost and complexity tend to cause barriers for information technology adoption in the healthcare industry. However, the significant aspects of quality and efficiency gains remain through the use of information communication and technology. Yet, access to these technologies is still deprived for many.

Most Indian metro hospitals, as well as small private clinics, are using the information technology for administrative purposes. Instead of accessing the medical records (birth to death) at any point of care by both patients and physicians is said to be a critical aspect of healthcare in these setups. Computerising the medical records of patients can improve the quality of Health and ICT the services.

Check Your Progress: 3

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. Explain the Health Belief Model.

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2. How can the Anderson theory and Transtheoretical model be explained based on skill acquisition?

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3. What are the barriers to Health Information Literacy?

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10.7 LET US SUM UP

In this Unit, we discussed the role of ICT in communicating health messages, how to use the Internet for health information, strategies and skill development for acquiring health information and major barriers and problems posed in assessing health information. Health communication is an exciting multifaceted area of applied communication inquiry that examines the role of messages, message exchange, and information about healthcare and health outcomes.

We also looked at how health information is imparted in various sectors of health. Dissemination occurs through a variety of channels, social contexts, and settings, -media, government organisations, community health workers and international organisations. This section includes advocacy and campaigns at the strategic level - how health information is working. Terminologies used explicitly in health communication, such as Health Literacy and Health Information Literacy, were explained.

Theoretical models which explain health information-seeking behaviour such as the Health Belief Model, Transtheoretical Model and Anderson's ACT theory were explained in order to understand the health information seeking behaviour. Even though these theories have adopted a cognitive approach for information gathering, it well defines the information-seeking capability of an individual. Barriers to health information seeking such as lack of information, problems in accessing, readability, specific social determinants and issues in India have been discussed. To sum up, it can be said that modern technology has in spite of the growth of healthcare and information, gained reach and access and the area is growing further.

10.8 KEYWORDS

Adaptive Control of Thought is a learning theory created by Canadian Psychologist John Anderson, that explains the process of acquiring knowledge and information through cognition. The main components of ACT are Modules, Buffers and Pattern Matcher.

Health Belief Model is the most widely accepted theoretical model to explain and predict changes in health behaviours. The model defines the key factors that influence health behaviours as an individual's perceived threat to sickness or disease (perceived susceptibility), the belief of consequence (perceived severity), potential positive benefits of action (perceived benefits), perceived barriers to action, exposure to factors that prompt action (cues to action), and confidence in the ability to succeed (self-efficacy).

Health Information Literacy: Health information Literacy (HIL) the set of abilities needed to recognise a health information need, identify likely information sources and use them to retrieve relevant information, assess the quality of the information and its applicability to a specific situation, and analyse, understand, and use the information to make right health decisions."

Health IT: The term "health information technology" (health IT) refers to the electronic systems being used by health care professionals – increasingly for patients which they use to store, share, and analyse health information.

Health Literacy: Health literacy(HL) is defined as the skills and competencies to which individuals have the capacity to obtain, process, and understand necessary health information and services needed to make appropriate health decisions.

Transtheoretical Model (TTM) (or Stages of Change Model) is a cyclic model developed by Prochaska and Diclemente (1983). TTM conceptualises behaviour change as a process involving a series of six distinct stages: pre-contemplation, contemplation, preparation, action, maintenance and termination. Each stage undergoes motivation to change, individuals apply cognitive, affective, and evaluation processes to progress forward. Each stage brings an individual closer to making or sustaining behavioural changes which are, in turn, explained as healthy behaviour change.

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10.10 CHECK YOUR PROGRESS QUESTIONS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Need for ICT in health - IT helps to bridge the information divide between health professionals and the communities they serve and between those involved in health research and the practitioners who need it. ICT acts as [1] To increase potential service and care and disease prevention. A delivery mechanism to poor and remote locations of a wide variety of services varying from improved public health education; [2] Easier transformation and reduced complexity; [3] Ensure safety and reduce difficulty in reducing risk behaviours; [4] Easier outcome for

intervention programmes and health delivery messages; and [5] Increase the transparency and efficiency of governance.

2. E-health finds applications in all sectors of health care and delivery which uses electronically possessed information. They are used by doctors, patients and Community Health Workers. Electronic health records (EHRs), electronic medical records(EMR), devices and technologies used to monitor and evaluate health care. Various mobile applications used to register and record health data, online services and training provided to doctors and health workers. Broader access to literature and internet technology for further reference and studies.

Check Your Progress: 2

1. Health literacy emphasis communication between health professionals and patients. People's knowledge, motivation and competences to access, understand, appraise and apply health information in order to make judgements and the decisions in everyday life concerning health care, disease prevention and health promotion to maintain or improve quality of life during the life course. Health information literacy originates from the field of library and information science and stresses individuals' role as active subjects in information acquisition rather than as objects of activities. Health information literacy has focused on information discovery.
2. Media plays a vital role in disseminating health information. Media informs and educates people, which ultimately aim to change the attitude & behaviour of the audience for achieving better health. Based on the reach and availability, different media forecast health information in different ways to society.

Check Your Progress: 3

1. Health Belief Model (HBM) identifies behavioural health change of an individual. The essential components of the HBM depend primarily on two variables (1) the value of a goal to an individual, and (2) the individual's appraisal of the likelihood of action in achieving that goal. The model defines the key factors that influence health behaviours as an individual's perceived threat to sickness or disease (perceived susceptibility), the belief of consequence (perceived severity), potential positive benefits of action (perceived benefits), perceived barriers to action, exposure to factors that prompt action (cues to action), and confidence in the ability to succeed (self-efficacy).
2. Theories and models pertain to health information and behavioural change shape health information seeking skill. The Internet, as online health care is enabling the growth of the practitioner-patient relationship and as well as of health workers and caretakers. The Adaptive Control of Thought is a learning theory created by Canadian Psychologist John Anderson. Explain the process and understanding and acquiring knowledge and information through cognition. The main components of ACT are Modules, Buffers and Pattern Matcher. TTM conceptualises

behaviour change as a process involving a series of six distinct stages: pre-contemplation, contemplation, preparation, action, maintenance and termination. These stages are transtheoretical and integrate principles of change from across a variety of theories of intervention.

3. Socio-Cultural disparity forms the main barriers in society, culture, language and facets of life that vary widely among socio-economic conditions. People with low levels of education, economic conditions and income. Credibility and trustworthiness of the source, poor information skill. Low readability of online health information and lack of interest also forms the barriers to health information.



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UNIT 11 ICT FOR DISABILITY

Structure

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11.1 INTRODUCTION

Technology has the potential for making significant impact in the lives of persons with disabilities. It can enable them to compensate for physical or

functional limitations and as a result allowing them to enhance their social and economic inclusion in society. Technology helps persons with disabilities to live an independent life to a great extent. Assistive technology provide greater independence to persons with disabilities by enabling them to perform tasks that earlier they were not able to accomplish or had much difficulty in doing.

Recent pandemic has forced the extensive use of technology by various people. As a matter of fact only technology is connecting all of us to the outside world and most of us have been using it to compensate for physical contact. Many organisation/companies have allowed the employees to work from home. Schools are teaching children through technology at their home, NGO's are providing counseling and other online support to the beneficiaries.

The rehabilitation services provided to persons with disabilities can be improved with the help of new advanced technologies. Keeping in mind the need and demand of persons with disabilities many new technological solutions have come into existence. The effective use of these technologies can enhance the functional capabilities of persons with disabilities and in turn they can contribute effectively for the society.

In this unit we will discuss about the various technologies useful for persons with disabilities.

11.1 LEARNING OUTCOMES

After going through this unit you will be able to:

- Describe the benefits and limitation of ICT for persons with disabilities;
- Analyse the emerging trends related to ICT for persons with disabilities;
- Discuss various types of ICT used at institutional level for persons with disabilities;
- Enumerate various types of technologies available for persons with disabilities for individual use.

11.2 ICT FOR PERSONS WITH DISABILITIES

Various Government, non-government organizations and individuals have taken initiatives for developing technologies after considering the needs of persons with disabilities. They have not only developed these technologies after research but they have also adapted several technologies developed in other countries to suit the local needs.

11.2.1 Terminologies

There are several terminologies that are used interchangeably for technology, let us discuss some of these terminologies.

11.2.1.1 Information Technology (IT)

Wikipedia definition of Information Technology (IT) is – “Information Technology the use of computers to store, retrieve, transmit and manipulate data or information in its various forms”.

11.2.1.2 Assistive Devices (AD)

The terms assistive device has been applied to a wide range of highly specialized mechanical, electronic and computer based consumer tools that are now commonly used in rehabilitation and education settings.

11.2.1.3 Assistive Technology (AT)

Assistive technology refers to any tool, item or product used for enhancing, preserving or increasing the competencies and abilities of individuals with disabilities to their potential.

11.2.1.4 Information and Communication Technology (ICT)

Information and Communication Technology are defined as diverse set of technological tools and resources used to communicate, and to create, disseminate, store and manage information. Basically ICT is technology that supports activities involving information such as gathering, processing, storing and presenting data. Increasingly these activities also involve collaboration and communication hence information technology has become ICT.

11.2.1.5 Universal Accessibility

Universal accessibility means making a device, product, website or a building usable by the larger number of people irrespective of their language, gender, height or abilities, bringing the concerns subject within the reach of maximum number of people.

11.2.2 History of Assistive Technology

Assistive Technology is a tool or a device that help persons with disabilities in doing their activities and it enhance their capabilities, it can be a simple device or a technological tool that can help persons with disabilities. If we look into the history of development of assistive technology we can say that in 1808 Pellegrino Turri invented a typewriter to help his childhood friend Contessa Fantion in the early stages of blindness so that he can write with the help of it. Since then there have been many other such attempts to strengthen the assistance for persons with disabilities. Louis Braille developed a language based on tactile writing system for the blind in 1921 this language is called Braille. Braille has made a great impact on the lives of persons with disabilities and it has helped them tremendously.

The invention of telephone was also a big contribution for helping individuals to communicate with each other from a long distance. This device was combined with a text telephone known as telecommunication device for the deaf that enabled them for communication with the people from long distance. Televisions, remote controls, talking calculator, talking

watches, speech recognition system etc. have made the life of persons with disabilities easier.

11.2.2.1 Benefits of Information and Communication Technology (ICT)

The use of ICT has brought enormous change in the life of persons with disabilities because of the enabling effect of technology now people talk about the potentials of persons with disabilities instead of their limitations and they are seen as productive partners in every segment of society. Some of the benefits of ICT are following

- i. Promotion of equal opportunities
- ii. Improved communication
- iii. Movement and mobility is increased
- iv. Participation in community is enhanced
- v. Increased independence in various spheres of life like home, school, college, community, markets and work place etc.
- vi. Persons with disabilities become self reliant
- vii. Participation of persons with disabilities in the various activities at home, school and work place is increased and the full potential of persons with disabilities is utilized
- viii. Persons with disabilities feel self motivated
- ix. Persons with disabilities feel more accountable
- x. Persons with disabilities get more opportunities, interaction and communication with others
- xi. Self respect and confidence of persons with disabilities is enhanced
- xii. Use of technology help in improving writing and organizational skills
- xiii. Technology help them in learning through their preferred mode that is i.e. visually or auditory
- xiv. Technology facilitates sharing of resources, expertise and learning material among the users
- xv. The accessibility is higher when technology is used
- xvi. Employment opportunities for persons with disabilities are increased by the use of technology
- xvii. Persons with disabilities can participate in sports and recreation with the help of technology

Check Your Progress - 1

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. Define Information and Communication Technology in your words.

.....

2. What is universal accessibility?

3. Write the names of any two technologies developed to assist persons with disabilities in early years.

4. List 5 benefits of ICT.

11.3 PRESENT AND FUTURE OF ICT

In India we have adopted the rights based approach after signing the UN convention on the Rights of Persons with Disabilities in 2006. The Rights of Persons with disabilities Act 2016 has compulsorily adopted this approach and emphasized that the rights of persons with disabilities should be protected and they should be treated with equity and equality in each sphere of life. Consequently it has been envisaged that all measures should be taken for full and effective participation of persons with disabilities at various places. We need to respect their individual differences and accept them and provide equal opportunities to live a healthy life. Technology in a great leveler for providing equal opportunities to persons with disabilities, as appropriate assistive technology enables them to work effectively. Persons

with visual impairment can read with the help of JAWS software, they can access the online library through it and the online material available in accessible format enables them to do their studies or work. Similarly there are other technologies available for persons with other disabilities. We will discuss about them later in this unit.

11.3.1 The Present Scenario

Development of ICT for all kind of disabilities is a huge task that needs research and innovations. In India around 70% of the population lives in rural areas that makes the task of providing technological intervention for such a diverse group of people very difficult. After the enactment of Rights of Persons with Disabilities Act 2016, 21 types of disabilities have been identified and most of the persons having these disabilities require assistive device for helping them in their daily living skills or vocational skills. Various initiatives taken by government or non government organisations for research and development of assistive devices have contributed for improving the quality of life of persons with disabilities. They have not only developed the technology for persons with disabilities but they have also been able to adapt the technologies developed in other countries to suit the local needs of India. These technological devices are provided by Government to the eligible persons without any cost under a scheme called ADIP scheme, apart from this these devices are available in the market too.

Children with intellectual disabilities quite often like to use computers provided the software in the computer is suited to their slow pace, they enjoy working on it and the technology enhances their capacity by gainfully engaging them through various tasks or games. To provide them instructions there are several games available which provide them simulations, demonstrations, problem solving and discovery learning.

Persons with visual impairment find the use of technology very convenient as they can read, write, walk or do various activities with the help of technology. There are various technologies available for them to help them in their mobility, education or daily living activities i.e. smart cane, screen magnifier, screen readers, Braille printers, Braille embossors, talking diary, talking calculator, talking ATM etc.

Telephone is one of the first technological devices developed for persons for hearing impairment. Alexander Graham Bell invented the telephone for helping his wife who had a hearing disability. Individual FM system, Loop induction system, speech processing software, computer assisted remote transcription, captioning and various types of alerting devices are available for persons with hearing impairment.

Similarly there are numerous other devices to help persons with locomotor disability and cerebral palsy. For persons with autism communication skill is very important and there are few devices available to assist them. Now-a-days app based support is also provided to persons with disabilities, recently Sense International India and Samsung have developed a mobile app named

Good Vibes for persons having deaf blindness to help them to communicate with their caregivers and family members.

11.3.2 The Future

In last few years there is a revolution in the area of technology. Artificial intelligence has been a key for this sector, but when a technology is developed very few companies think about the accessibility of this technology to persons with disabilities. It is high time that as a country we think about universal accessibility of the technologies being developed for various areas of life, inclusiveness of the technologies will enhance the capabilities of persons with disabilities to a great extent. If we take the example of persons with deafblindness there is hardly any device developed in India for supporting them. The persons with deafblindness have a unique condition – a combination of visual and hearing impairment. Although the degree of deafness or blindness varies, the combination of dual sensory loss leads to unique problems in an individual's capability for communication, mobility and their ability to access information. The technological intervention with the help of an app can provide them ability to communicate with their care givers and family members. Keeping this in mind recently Sense International India joined hands with Samsung to develop a communication tool for the deaf blind. This is a two way communication app that allows the deaf blind to send and receive messages to friends, family or anybody else through their smart phone. More such initiatives needs to be taken to meet the needs of persons having various disabilities.

Recently the whole world has witnessed COVID 19 Pandemic and it has resulted that all the educational institutions are providing educational support to their students with the help of ICT. Mobile phones have been used extensively, the teachers are using various platforms to teach the children from their home. Although our educational system was not prepared to teach the students through mobiles or computers but despite all the difficulties they have been doing this for last few months. In this situation it is very difficult for persons with disabilities and their families to cope up with the lack of accessibility to suit the individual needs. This is a right time when as a country we think about the various needs of children/individual with or without disabilities, the pedagogy also needs to be aligned to suit the needs of all children.

There is a need to emphasise more on massive online open courseware (MOOCS) that can enable person with or without disability to learn at their own pace or place. Gaming has also be emerged as a powerful tool. Judicious use of accessible gaming will help the persons with disabilities for leisure, personalised virtual teacher, automated assessment and evaluation etc. Mobile technology is also helping the persons with visual impairment, hearing impairment and other disabilities to a large extent, more such initiatives needs to be taken for developing universally accessible technologies for persons with disabilities.

Check Your Progress – 2

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. Rights based approach means

.....
.....

2. Good Vibes app is developed by and
To help individuals with

3. Two technological devices beneficial for persons with visual impairment
are and

4. Loop induction system is beneficial for persons with

11.4 ICT FOR VARIOUS TYPES OF DISABILITIES

ICT is used for providing support to persons with disabilities, when it is used in educational institutions following the principles of universal accessibility then it helps students with disabilities tremendously and they are included in all the activities of that particular institution. Similarly when a website is designed following the universal accessibility then the needs of persons with disabilities are ensured. If at the time of planning of a course/ lecture/ discussion /visual aids/video tapes/printed material and practical/field work the accessibility for students with disabilities are kept in mind then it help in their inclusion and learning to a great extent. Distance learning programmes that incorporate universal accessibility features helps the students with diverse needs to learn at their own pace.

Use of technology for providing accessible education to students with disabilities is being done at institutions level as well as personal level. Means the technology is used by a bigger system i.e. university or a college to support the educational needs of students enrolled in various programmes including students with disabilities. On other hand the technology that helps an individual to compensate for his or her limitation or restriction in one or more areas is used at individual level. Let us discuss about this in brief.

11.4.1 Technology Used at Institution Level for Instruction

With the help of technology education can be more open and accessible to all the students enrolled in a particular programme specially the students with disabilities. With the help of these technologies support can be provided to the students with disabilities in their teaching learning process.

11.4.1.1 Online Education

Online education is very convenient for the students with disabilities as they can learn from the convenience of their home and use of accessible softwares

enables them to compensate for their inability. In online programmes there is a scope to learn at a flexible time and pace. If the students with disabilities are provided the content in accessible format then they benefit a lot from it.

11.4.1.2 Radio Programmes

Radio Programmes are being used by various educational institutes to provide education. Some of the institutions have their own community radio channels whereas other has access to FM channels or Gyanvani channels. The students with disabilities benefit from these programmes enormously. IGNOU has Gyanvani programme that offers interactive radio counseling to the students which enables them to interact with their teachers at the ease of their home. Radio programmes are not accessible to the students having hearing impairment as they will not be able to hear the discussions. The students with speech impairment will also not be able to participate in this because of their speech problem. Therefore while planning the educational support for a diverse group of learners, alternative mode for supporting students with other disabilities should be considered. In this case the students with speech and hearing impairment can be provided facility for live chats or support through telecommunication devices to the deaf students.

11.4.1.3 Teleconferencing

Teleconferencing facilities enrich the learning experiences of all students enrolled in any educational programme provided the needs of students with disabilities are considered before planning these programmes. To suit the needs of diverse group of learners, the teleconferencing programme can be made in universally accessible formats wherein the captioning and sign language interpretation is inbuilt into a programme along with the live discussions and powerpoint presentations.

11.4.1.4 Web Conferencing

Web conferencing can also be planned by adding sign language interpretation, captioning and audio recordings or descriptive captions/descriptive narrations etc. to suit the needs of individuals with various disabilities.

11.4.1.5 Mobile Phone Based Education

Smart phones have become an integral part of everyone's life in this technological era. Judicious use of mobile application can go a long way in providing quality learning experiences to students with disabilities as they can learn from the mobile application based contents provided to them by their educational institute. IGNOU has also developed a mobile application 'IGNOU e-Content' to enable the learners to access its learning material through their mobile phones. The learners can access their course material from anywhere at their convenience.

11.4.1.6 E-Resources

Availability of e-resources in accessible form are very beneficial to students with disabilities, Accessible libraries having content in e-Pub format or in

other formats suitable for persons with disabilities are providing enormous support to students with disabilities i.e. Sugamya Pustakalaya and Bookshare. IGNOU has also developed a digital educational resources repository of self learning materials of its programmes. The repository named as e-Gyankosh has all the instructional material and video programmes developed by IGNOU and it provides open access to all. E-Gyankosh is very popular among the students with disabilities irrespective of their enrolment in IGNOU.

11.4.1.7 Television

Television is very powerful tool for providing education to the learners with disabilities at their doorstep. There are several educational channels that provide facilities for learning in different subjects ranging from primary to higher education level. Gyan Dharshan is one of such channel in IGNOU that provides a variety of programmes across all disciplines to cater needs of wide range of viewers in higher education. It is beneficial for the learners of formal education system as well as other stakeholders. Learners with disabilities find it very useful for their studies.

11.4.2 Technology used at Individual Level

Persons with disabilities use various technologies to accomplish various tasks as per their need they are discussed here for the benefit of our learners.

11.4.2.1 Technology for Persons with Intellectual Disability

Technology can compensate for the functional limitations of an individual. Persons with intellectual disability should be introduced to the technology as early as possible. The teachers need to provide them these devices in all settings including home, school, work and recreation. Because these children have limitation in their mental functioning therefore there should be a consistency in the kind of technology available, how it is used and instructional method for operating the particular device. When a child is being taught a new technology their previous learning needs to be considered. The purpose of instruction is to provide them interactive experiences to promote learning among them. Several instructional programme for children with intellectual disability are drill and practice, instructional games, tutorials, simulations, demonstrations, problem solving, discovery learning etc.

Usually the persons with intellectual disabilities are provided instructions with the help of technology. There are several technologies for daily living skills for persons with intellectual disabilities like electronic planners remind them about the various tasks that they need to do at a particular time. When the child feel lonely they can be given the talking doll or any other toy. Soothers are objects that help them to keep calm and engaged i.e. electronic devices that can make sounds or anything as simple as a small soft toy or a small pebble that can be kept in the pocket. Video based instructional system can help them in learning functional life skills such as grocery shopping, writing a cheque or using ATM etc.

Technology for assessment of children with intellectual disability has also been developed wherein integrated assessment in detail and records of each child can be kept in the assessment tool. Media Lab Asia in collaboration with C-dac has developed (Punarjjani) a software tool to equip the teachers for the progress assessment and evaluation of children with intellectual disabilities.

11.4.2.2 Technology for persons with Locomotor Impairment and Cerebral Palsy

Cerebral Palsy is a group of non-progressive condition of a person characterized with abnormal motor control posture resulting from brain injuries incurred during the prenatal parinatal or infant period of development. Due to problem in movement these children face many problems in their life. Many persons with cerebral palsy have slight impairment whereas others have serious problems that cause them to have difficulty in performing even the most common task. Technological intervention will helps them in overcoming these problems and live independently as far as possible.

Persons with Cerebral Palsy with mobility impairment will need devices that can help them to go around they can use these devices for their studies or workplace. Wheelchair is the main device for mobility of children with cerebral palsy or locomotor disability. Now-a-days electronic wheelchairs are available at reasonable prices which are better than the manual wheelchairs. In case the individual is not able to use the wheel chairs they can be modified with a provision of switches that come in a wide variety. Sometimes they are shaped like a button, hand or foot operated or operated by the waving of hand etc. These switches can be selected depending upon the condition of the individual.

The most common form of human communication is speech. Children with Cerebral Palsy in some cases are totally unable to speak or in some cases they are not able to communicate effectively with their speech. In this case they may need a non speech mode of communication as an alternative of spoken language. They need alternative mode of communication therefore a few alternative and augmentative communication (AAC) devices have been developed for use of non speech mode to support these students. AAC is an integrated group of components including the symbols, aids, strategies and techniques used by individuals to enhance communication. In this system multiple components for communication are available. Some of these devices have been developed indigenously whereas others are adopted from other countries. The devices developed in foreign countries are - dedicated communication device, the alpha talker, the walker talker, the digital augmentative communicator, canon communicator, vocaid, board maker software, whereas Sanyog, Gup Shup, pictorial communication software, aditi, idraw, voice output communication aid are developed in India. These devices assist children with limited communication or no communication to communicate with their families and care givers effectively.

There are several other devices that help children with cerebral palsy and locomotor disability in their daily living activities these are computers, mobile phones, voice recognition, switches, adaptive key boards, touch sensitive screens, infra red sensors with pneumatic switches.

11.4.2.3 Technology for Persons with Hearing Impairment

Persons with Hearing Impairment are those who are having a hearing loss and because of that they are not able to process linguistic information through auditory channels with or without amplification. They may be able to hear some sounds but may not be able to distinguish words. Some other people may not be able to hear sounds at all. For these people prompts such as alarm, beep and spoken messages can be an issue of concern as they will not be able to hear. The devices that help to enhance a person's hearing capacity is called an assistive listening device. Apart from these devices telecommunication devices are developed to assist the person with severe hearing impairment. As mentioned earlier first technological device designed for persons for hearing impairment is telephone that was originally invented by Alexander Graham Bell for helping his wife who had a hearing disability. The advancement of technology has led to the development of a wide range of modern assistive listening and telecommunication devices that assist students with hearing impairment to participate more effectively in various areas of life.

Telecommunication devices allows the users to use a keyboard to type and receive messages over the phone. Mobile phones are widely used for communication through SMS or other forms of messages. For education of persons with hearing impairment individual FM system is used, it is an assistive listening device that improves the sound noise ratio for one person by using a remote microphone placed near the source of sound. A basic FM system consists of two units, a transmitter and a receiver and is like having an individual radio station that transmit and receive on a single frequency. These systems are useful in classroom lecture, in a restaurant, theaters, public meeting place, corporate conference rooms and other large areas for gathering.

There are sound enhancement technology devices that transmit the sounds from the source to the receiver by improving the signal to noise ratio. These are designed to deliver the sound directly to the listener who has direct control over intensity of sound. This technology is used to enhance the auditory signal during communication in the group, in one to one situation or audio video signals. Some examples of sound enhancement devices are FM systems, infrared system, hardware system, and induction loop system etc.

Hearing aid is an amplifier which helps the person to listen to a sound with minimum distortion by increasing the intensity of sound. The role of hearing aid is to increase the level of sound so that person with hearing impairment can detect and make use of auditory signals. We can say that hearing aid is an electronic device that amplifies sound to a degree and in a manner that will enable a person with hearing impairment to utilize remaining hearing

capacity effectively. The hearing aids are of two types one is individual hearing aid and another is group hearing aid.

There are technologies available for assessment of hearing impairment in the audio lab. The software packages allows the users to record the speech signal and graphically added the same to prepare the test words. Some of the devices used for assessment are speech processing software, visual reinforcement audiometry, pure tone audiometers, speech audiometry, speech trainer and visual aids.

There are several devices available for providing support services for daily life for persons with hearing impairment. Computer assisted remote transcription allows a stenographer to transcribe a meeting from a remote location. The facility of captioning means addition of text to a visual display that helps in conversion of spoken words into text. In many programmes on television facility of captioning is available, live speech captioning is another type of captioning that provides accessibility of lives programmes and speeches to the persons having hearing impairment. In educational setting this technology is very beneficial for students having hearing impairment as the speech is typed simultaneously by a stenographer and the text will be displayed on a screen. Alerting devices are those that listen to the sounds on behalf of the persons with hearing impairment and draw their attention to the sound source by generating a visual signal. Example of this is a door bell that makes low sound and a flash light, similarly in telephone vibration or flash light is used.

Telecommunication devices for deaf are widely used by them to make or receive a phone call. These devices generally attached with the telephones and they have a small key board for typing and a screen to display an incoming or outgoing messages. These devices can be attached with a printer to have a copy of conversation. To use the telecommunication devices the users have to type the messages that they want to convey, the written text will be automatically converted into speech and will be transmitted over the phone to another user. It converts the messages back into text form. These technologies are helpful for students with hearing impairment to interact with each other and other stakeholders.

11.4.2.4 Technology for Persons with Visual Impairment

ICT advancement has increased the quality of life of persons with visual impairment and their education is significantly enhanced by the application of technology. The modern technology has given amazing experiences to persons with visual impairment understand the unseen world through non-visual media. In this digital era mobile phones, computers, i-pad, scanner and internet has provided them opportunities for flying out of their boundaries and interact with the world. They can obtain information on any topic with the help of internet which was earlier difficult for them to get either in Braille or from other sources. Availability of e-books, e-newspapers and e-resources have helped them tremendously.

To reduce the limitations for students with disabilities several efforts are required like curriculum adaptation, training in plus curriculum, providing

modified teaching learning materials and access to technologies suitable for them. There are several technologies that help them in mobility and they can go at the desired places without any dependence on others, some of these are - white cane fitted with laser, sonic guide, touch traffic light signal, path sounder, mowat sensor, mobility assisting softwares and smart cane. All these devices help persons with visual impairments in moving around independently.

There are several technological solutions that help persons with visual impairment in their education these are - close circuit television, large print computer, screen magnifiers, screen readers, Braille embossers, referable Braille displays, optical character recognition, e-books, audio books, digitally accessible information system (DAISY), text browsers, voice browsers, audio devices, Braille note takers, talking calculator, electronic Braille, internet, mobile phones etc.

There are numerous technological devices that help persons with visual impairment for their life skills these are – Braille printing system, handy Braille, talking diary, talking watch, talking tool box, talking alarm clock, talking color detector, talking measuring tap, talking kitchen scale, talking microwave oven, talking ATM etc. With the help of these devices persons with visual impairment can undertake their daily living activities at home, workplace, market, recreation centres and in community at large. They do not need to be dependent on others for their routine activities related to various areas. The need is to provide them proper orientation and mobility training initially so that they are well aware about their surroundings and the places where they need to go frequently. The devices mentioned above will help them tremendously in various settings.

11.4.2.5 Technology for Persons with Autism

Technology plays a very important role for the benefit of persons with autism. Autism is a developmental disorder characterise by triad of impairments in social interaction, verbal and non verbal communication and repetitive interests or pattern of behavior. It is developmental disability with onset in early years and causing serious problems or delays in different areas of development of children. They are not able to relate to others and may show abnormal interests in objects, may not maintain eye contact with others, remain aloof and asocial, resist change and insist in sameness.

For children with autism intervention needs to be provided for enhancing their verbal and non-verbal communication, social interaction, leisure and play activities for their holistic development. With the help of technology they can become participative and enhance their adaptive skills. Devices like talk pad, language master, voice output communication aid, audio taping, sanyog, autism my voice communicator, first then visual schedule, my pictures talk, pictello, speech journal etc. are devices that help them in communication with their environment.

A person having autism find difficulty in focusing attention as a result learning become difficult for them. Use of technology helps them in learning as their often interested and attentive to the videos and use of computers.

Selection of appropriate technology will help in enhancement of their academic achievements, a few technology based devices are – video tapping, computers, LCD projector, autism bundle etc. there are various other softwares available for them for educational purposes.

The persons with autism have difficulty in social interaction and they show less attention to social stimuli, smile and do not look directly at other person. They have very less or no eye contact, use of appropriate technology can help them to enhance their social skills, a few of them are – bigmack, audio tapping, visual impact, symtrend ADL, tool factory beep, injini child development game suite, sound beginnings and touch it transport etc.

Check Your Progress – 3

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. How educational institutions can provide instructions to persons with disabilities by using technology?

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2. Write any two benefits of mobile phones for education of persons with disabilities.

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3. Name any two devices used by persons with hearing impairment?

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4. How video based instructional system will help persons with intellectual disability?

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11.5 LET US SOME UP

In this unit we discussed about the various technologies used for persons with disabilities. We have discussed various terminologies that are used interchangeably for technology. The benefits of technology were also discussed in this unit.

We traced the history of assistive technologies and discussed the present scenario regarding the technologies in the country. We also discussed the future of technology for our country.

Technologies are beneficial for persons with disabilities and these are being used by various educational institutions to provide effective support to students. We discussed the various types of technologies that the being used by various institutions for providing instructions to the students and how they can help students with disabilities.

There are several technologies that help persons with disabilities individually to accomplish various tasks. We discussed about such technologies available for persons with intellectual disability, persons with locomotor disability and cerebral palsy, persons with hearing impairment, persons with visual impairment and person with autism.

To some up we can say that technology plays very important role in the life of persons with disabilities and it offers immense possibilities for their education and daily living skills. Use of appropriate technology can empower the persons with disabilities and they can become the productive part of society.

11.6 REFERENCES AND FURTHER READINGS

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11.7 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Information and Communication Technology can be defined as a set of technological tools that are used to communicate, create, spread, store and manage the information.
2. Universal Accessibility means a material website or building is made in way that can be used by a large no. of people without any hindrance and it will be equally useful for all
3. Typewriter and telephone.
4. Promotion of equal opportunities, improved communication, enhanced community participation, persons with disabilities become self reliant confidence of persons with disabilities increase.

Check Your Progress: 2

1. Persons with disabilities are treated with equity and equality in each sphere of life and their individual differences are respected
2. Sense International India and Samsung, deafblindness
3. Smart cane, Screen Readers
4. Hearing Impairment

Check Your Progress: 3

1. Educational Institutions can provide instructions to students with disabilities very effectively through the use of radio programmes, mobile apps, web conferencing , e-resources, teleconferencing and television programmes etc.
2.
 - i) Study material can be accessed from anywhere
 - ii) Students can be provided continuous support by the teacher through smart phones
3. Hearing aid and induction loop system
4. It will provide them interactive experience and will enhance their functional limitations to some extent.

